

Element E: Application of STEM Principles

STEM concepts applied:

- Physics: Buoyancy (ρVg), drag force ($F_d = 0.5\rho v^2 C_d A$), and torque balancing on thruster mounts
- Engineering: Structural load vs. weight displacement tradeoffs, waterproof housing design
- Technology: PS/2 protocol logic for motor mapping, Arduino PWM output regulation
- Math: Measurement error margin calculations, tether weight offset prediction

Reviewed by Judge Ned Vorhees (expert in marine robotics), who validated buoyancy plans and gave feedback on cable drag mitigation. Recommendations implemented pre-competition.